

Probability and Statistics. Week 1

1. (Walpole 1.2) The reaction time, in milliseconds, of 20 amateur racers is measured. A sample is as follows:

187.1	214.1	207.2	218.1	192.9	224.3	201.7
237.1	194.4	205.0	189.2	203.3	230.0	228.5
192.5	217.7	221.1	197.7	180.4	211.2	

- (a) Calculate the sample mean and sample median for the above sample values.
 - (b) Compute the 10% trimmed mean.
 - (c) Do a dot plot of the reaction time data.
 - (d) Using only the values of the mean, median, and trimmed mean, do you have evidence of outliers in the data?
2. (Walpole 1.5 and 1.11) The effect of regular exercising on a cholesterol level is being evaluated. All participants are divided into two equal groups. Reductions in cholesterol levels reported below.

Control group:	7	3	-4	14	2
	5	22	-7	9	5
Treatment group:	-6	5	9	4	4
	12	37	5	3	3

- (a) Do a dot plot of the data for both groups on the same graph.
 - (b) Compute the mean, median, and 10% trimmed mean for both groups.
 - (c) Explain why the difference in means suggests one conclusion about the effect of the regimen, while the difference in medians or trimmed means suggests a different conclusion.
 - (d) Compute the sample variance and the sample standard deviation for both groups.
3. (Walpole 1.13) A manufacturer of electronic components is interested in determining the lifetime of a certain type of battery. A sample, in hours of life, is as follows:

123, 116, 122, 110, 175, 126, 125, 111, 118, 117

- (a) Find the sample mean and median.
 - (b) What feature in this data set is responsible for the substantial difference between the two? Recalculate the mean without that feature and compare the results.
4. (Walpole 1.17) A study of the effects of smoking on sleep patterns is conducted. The measure observed is the time, in minutes, that it takes to fall asleep. These data are obtained:

Smokers (A):	69.3	56.0	22.1	47.6
	53.2	48.1	52.7	34.4
	60.2	43.8	23.2	13.8
Nonsmokers (B):	28.6	25.1	26.4	34.9
	29.8	28.4	38.5	30.2
	30.6	31.8	41.6	21.1
	36.0	37.9	13.9	

- (a) Find the sample mean for each group.
 - (b) Find the sample standard deviation for each group.
 - (c) Make a dot plot of the data sets A and B on the same line.
 - (d) Comment on what kind of impact smoking appears to have on the time required to fall asleep.
5. (Walpole 1.18) The following scores represent the final examination grades for an elementary statistics course:

23	60	79	32	57	74	52	70	82
36	80	77	81	95	41	65	92	85
55	76	52	10	64	75	78	25	80
98	81	67	41	71	83	54	64	72
88	62	74	43	60	78	89	76	84
48	84	90	15	79	34	67	17	82
69	74	63	80	85	61			

- (b) Construct a relative frequency histogram, and discuss the skewness of the distribution.
 - (c) Compute the sample mean, sample median, and sample standard deviation.
6. (Walpole 1.20) The following data represents the length of life, in seconds, of 50 fruit flies subject to a new spray in a controlled laboratory experiment:

17	20	10	9	23	13	12	19	18	24
12	14	6	9	13	6	7	10	13	7
16	18	8	13	3	32	9	7	10	11
13	7	18	7	10	4	27	19	16	8
7	10	5	14	15	10	9	6	7	15

- (b) Set up a relative frequency distribution.
- (c) Construct a relative frequency histogram.
- (d) Find the median.

7. Implement in code:

- Mean computation;
- Median computation;
- Standard deviation;
- MAD computation;
- Histogram plotting.

You can download required ipynb notebook and data sample from Moodle.

Note: you can use Google Colab or configure local Jupyter notebook environment.

8. Rolling a fair die you have equal chances to observe 1, 2, 3, 4, 5, or 6.

In case you roll 2 dice, resulting sum is between 2 and 12. Both 9 and 10 can be obtained in two ways:
 $9 = 3 + 6 = 4 + 5$, $10 = 4 + 6 = 5 + 5$.

In case you roll 3 dice, both 9 and 10 obtained in 6 ways.

Then, why 9 shows is more likely to be observed when you roll 2 dice, and 10 is more frequently observed when you roll 3 dice?

Note: word "die" is a singular form of "dice". However, nowadays those "dice" is used both as a singular and plural.

9. When you roll a die 4 times probability that 1 shows up at least once exceeds $1/2$. However, when you roll 2 dice 24 times, probability that 1 appears on both dice simultaneously at least once is below $1/2$. This result is contradictory, because probability of observing 1 is 6 times more than probability of observing two 1's, and 24 rolls is exactly 6 times more than 4 rolls. Explain this contradiction.
10. We flip 3 fair coins. At least two of them show the same side, and head and tails on a 3rd coin are equally likely to appear. It follows that all three show the same side is $1/2$. Is it true?